claim	our assessment	rests on
D answering, not raw prediction	true in comparative form; not a new inference principle; survives the objection from statistical practice	**????????** and Props. 1.3 to 1.6
S a logical theory, not a program	descriptive part true of the declarative presentations of fundamental theories (an observation); explanatory part open, and relative to a reasoner	§ 1.4; **????????**
E explaining propositions	true in the unification sense, with at least four gaps: asymmetry, approximate correction, irrelevance, conjunction	Props. 1.7 and 1.8 and **????**
L the ladder	SAI is on step 2, with partial results on step 3; step 4 has no formal handle	**??????**
P progress over raw-data prediction	replace by a list of formal differences: advantages, neutral differences, liabilities; none concerns the shape of hypotheses in the idealised setting	Tab. 2

Table 1: The argument map: our assessment of five claims of the axiom view (arguments), and the results each rests on (theorems in their stated settings).

1 What a theory does and what it is

What does a scientific theory do? It may predict observations, the view behind Solomonoff induction (Solomonoff, 1964); answer questions posed in some language, why-questions included (van Fraassen, 1980, ch. 5); explain (Hempel and Oppenheim, 1948); or solve problems (Laudan, 1977). And what is it? A program that prints or predicts observations; a set of sentences closed under consequence (Suppe, 1977); a class of models (Suppes, 1960; van Fraassen, 1980); a core of models with intended applications (Sneed, 1971; Balzer et al., 1987); or a practice of exemplars and skills (Kuhn, 1970, 1977; Polanyi, 1958). The axiom view says that a theory answers questions posed in a given language and is a logical theory; the raw-data view, that it predicts observations and is a program. This section assesses the axiom view against the theorems of this paper. Its propositions only make claims precise enough to test (proofs in Appendix A); the rest is argument, and we say where an argument rests on a theorem.

1.1 An argument map

We separate five claims. **D**: a theory answers questions posed in a given language rather than predicting raw data; we read it comparatively, as a step away from raw-data prediction, not the end point. **S**: a theory is a logical theory, not a program. **E**: theories explain propositions. **L**: the ladder of **??** (raw data; questions posed with existing abstractions; theories that extend the language; theories that revise the base abstractions). **P**: SAI is progress over raw-data prediction, both in what a theory does and in what it is. Table 1 gives our assessments.

What rests on theorems. Proved, in the settings stated: the idealised formalism cannot see the difference between a theory and a program, in representation, vocabulary, reasoner or formulation (Section 1.4); under a cost on the written theory, usefulness depends exponentially on the pair (reasoner, vocabulary), and on declarative against procedural form only by a constant factor once intermediate concepts are available and the reasoner can chain through them (in the model of **??**; open beyond it); theoretical vocabulary is dispensable in the ideal mixture, while relations or entities pay for finite theories and under charged deduction (**??**); answering questions differs from raw-data prediction in target, data type, coherence and partiality, not in

the inference principle (Section 1.3); "Kepler's laws are only approximately right" is a limitation of proof-requiring compatibility with deterministic theories, not of the axiom view as such (**??**, with computations on real residuals within stated pools). Computed, conditional on the pool of forms and the codes: simplicity among compatible forms is relative to the language on noise-free interval data, and with graded fit the data decide (**??**). Proved, with an identification that is an argument: Hempel's inductive-systematisation reply fails in the idealised Bayesian setting (**????**). Arguments: the descriptive observation; why declarative shape is good (open); that D survives the objection from statistical practice; the general claims about explanation (the toys are proved); that the axiom view models the declarative core of theories and misses their procedural, statistical and dynamic sides (**??**); the recommendation to say nothing realist about induced theories, which rests on a possibility result (**??**); and the readings of the theorems as answers to D, S and E.

The ladder. We agree with L. The base language of step 2 may be highly theoretical: heliocentric longitudes presuppose Copernican geometry. No theory-neutral observation language is claimed, so the classical objections to one do not apply in the same way; but theory-ladenness (Hanson, 1958) still biases comparisons between theories that disagree about the base language, since a theory that revises the base abstractions is scored in a language it rejects. This is why step 4 is listed and SAI is placed on step 2.

1.2 The descriptive observation and the explanatory question

Claim S combines a description with an explanation, which we keep apart. *The observation:* textbook presentations of fundamental theories are declarative. They state laws, definitions, principles, and models given by equations over mathematical structures; they are not procedures that print predictions. This is an observation about texts and needs no theory of why it holds. Two qualifications. *Declarative is not sentential:* physics texts write something closer to a class of models specified by equations, in the manner of Suppes's set-theoretic predicates and models of data (Suppes, 1960, 1962), than to first-order axioms. We use sentences because questions and data are sentences and the prior of SAI is a prior over texts; the question-answering view itself is neutral between sentences and classes of models, since in partition semantics a question is a partition of logical space and a theory answers it when its models lie in one cell (cf. Groenendijk and Stokhof, 1984). *Theories in use have procedural parts:* methods attached to concepts, numerical schemes, perturbation series, simulation codes. In quantum chemistry a "level of theory" is a computational method, and even one position of Mars from Kepler's first two laws needs root-finding for Kepler's equation. So the observation concerns textbook presentations, not theories in all their uses.

The question, why the declarative shape is good and whether an ideal inducer should produce it, is open. Craig's construction rules out an answer based only on what can be represented (Section 1.4), so an answer must be relative to a reasoner with costs, or to a community of reasoners. In the model of **??**, the best formal answer we have, what matters is a reasoner that can do algebra or chain through intermediate concepts, together with those concepts, and not logical form as such. The explanation may lie in communication and piecemeal checking, for which there is no formal setting yet.

Question 1.1 (Why declarative shape?)**.** Is there a formal setting for a community of bounded reasoners who communicate, check and extend theories piece by piece, in which theories of declarative shape provably do better than programs?

A natural argument for S is that a theory answers in many directions (Kepler's third law gives the period from the semi-major axis and the semi-major axis from the period; **??**), while a program computes one direction cheaply. The second half holds of particular programs, not of programs: a generic solver applied to a declarative law costs $O(1)$ extra length, a program with internal variables answers every direction of a parity law in $O(m)$ operations, and under program length directions cost nothing (**????**). Only for a fixed weak reasoner and a cost on the written theory do directions matter: a direction-complete rule base for one parity law needs exactly m times the clauses of the law, m the number of directions, without intermediate concepts or with one round of reasoning, and at most 9 times its written length with intermediate concepts and unbounded rounds (**??**).

1.3 What a theory does: questions against raw data

A question is a finite set $\mathcal{Q} = \{\varphi_1, \dots, \varphi_k\}$ of pairwise B-incompatible sentences whose disjunction B proves (**??**); $[k] := \{1, \dots, k\}$. Following the logic of questions (cf. Belnap and Steel, 1976; Hintikka, 1999):

Definition 1.2 (Answers)**.** A theory T *answers* $\mathcal{Q}$ if $B \cup T \vdash \varphi_i$ for some i, and *answers it partially* if $B \cup T \vdash \bigvee_{i \in J} \varphi_i$ for some J with $\emptyset \neq J \subsetneq [k]$.

In the Kepler example (**??**) the data are interval sentences, and questions are covers by them ("where is Mars at time t, to within $2'$?") or structural ("is the orbit of b planar?"). Kepler's laws without the elements of the orbits answer structural questions and entail no sharp position datum (**??**); with the elements they answer position questions too. A raw-data predictor answers one question, "what is the next observation?", and answers it fully. SAI answers every question the base language can pose and reports how much of its posterior answers none.

Proposition 1.3 (The answer distribution)**.** *Let $M_A(D) > 0$, let $\mathcal{Q} = \{\varphi_1, \dots, \varphi_k\}$ be a question, and let $\mathrm{Un}_D(\mathcal{Q}) := \mathbb{U}(E_D \setminus \bigcup_i O_{\varphi_i})/\mathbb{U}(E_D)$ be the posterior probability that the random theory proves no direct answer. Then (a) $\sum_{i \le k} \mathrm{Bel}_D(\varphi_i) + \mathrm{Un}_D(\mathcal{Q}) = 1$; (b) for $\emptyset \neq J \subsetneq [k]$, $\mathrm{Bel}_D(\bigvee_{i \in J} \varphi_i) \geq \sum_{i \in J} \mathrm{Bel}_D(\varphi_i)$, the difference being the posterior probability that the random theory answers $\mathcal{Q}$ partially through J but proves no φ_i with $i \in J$, and for $J = [k]$, $\mathrm{Bel}_D(\bigvee_i \varphi_i) = 1$ and the difference is $\mathrm{Un}_D(\mathcal{Q})$; (c) for $\mathcal{Q} = \{\varphi, \neg\varphi\}$, (a) is the trichotomy $\mathrm{Bel}_D + \mathrm{Dis}_D + \mathrm{Indep}_D = 1$ of* **??**.

On E_D the random theory is consistent, so the events "proves φ_i" and "proves no answer" partition E_D (Appendix A). In SOI, whose hypotheses are labellers of sentences, the analogue of Un_D mixes logical independence with a labeller's silence (**??**), and the answer masses need not respect the incompatibility of the answers unless the labeller is consistent.

The conservation-law world. What is a law worth to each task? For even m, micro-states are $x \in \{0,1\}^m$ with weight $|x|$; the dynamics σ is a uniformly random permutation of each weight class, so the weight is conserved (the law); x_0 is uniform on the class of weight $m/2$, of size $Z := \binom{m}{m/2}$, and $x_{t+1} := \sigma(x_t)$. Raw data are the trajectory; questions are "what is $|x_t|$?". The raw predictors Raw^+ and Raw^- are Bayes for "σ is a uniformly random permutation of each weight class", respectively "of $\{0,1\}^m$"; the question predictors Qa^+, Qa^- and $\mathsf{Qa}^{\mathrm{bin}}$ are Bayes for "$|x_t| = |x_0|$ with $|x_0|$ uniform on $\{0, \dots, m\}$", for "the $|x_t|$ are i.i.d. uniform on $\{0, \dots, m\}$", and for "the $|x_t|$ are i.i.d. binomial$(m, \frac{1}{2})$"; Mix gives prior $\frac{1}{2}$ to each of the two hypotheses of a task. Let t^* be the length of the cycle of σ through x_0; the step-t loss is that of predicting x_{t+1} from $x_0, \dots, x_t$.

Proposition 1.4 (The conservation-law world)**.** *(a) For $t < t^*$ the step-t losses of* Raw^+ *and* Raw^- *are* $\log(Z-t)$ *and* $\log(2^m-t)$*, and for $t \geq t^*$ both are* 0*. The law's gain $\kappa_t := \log \frac{2^m-t}{Z-t}$ ($t < t^*$) is nondecreasing in t; for $t \leq Z/2$ it satisfies $\log \frac{2^m}{Z} \leq \kappa_t \leq \log \frac{2^m}{Z}+1$, where $\log \frac{2^m}{Z} = \frac{1}{2}\log \frac{\pi m}{2}+o(1)$, and its share $\kappa_t/\log(2^m-t)$ of the step loss is $O(\log m/m)$. (b)* Qa^+ *pays* $\log(m+1)$ *for $|x_0|$ and* 0 *afterwards,* Qa^- *pays* $\log(m+1)$ *per question; over $t_{\max}+1$ questions the law's share of the no-law loss is $t_{\max}/(t_{\max}+1)$. (c) In each task* Mix *pays at most one bit more than the better of its two hypotheses, on every trajectory. (d) For $t < t^*$, $\log(2^m-t) = \kappa_t + \log(Z-t)$ with $\kappa_t = -\log P_{\mathsf{Raw}^-}(|x_{t+1}| = m/2 \mid x_0, \ldots, x_t)$: the law's raw gain is exactly the loss of* Raw^-*'s own answer to "what is $|x_{t+1}|$?". The loss of* $\mathsf{Qa}^{\mathrm{bin}}$ *on each question is κ_0.*

The proof (Appendix A) uses that under a uniformly random permutation the next state, given a path of distinct states, is uniform on the unvisited states and the starting state; the rest is Stirling's formula and, for (c), $-\log(\frac{1}{2}2^{-a}+\frac{1}{2}2^{-b}) \leq \min(a,b)+1$. A computation (code in the accompanying repository) gives, at the start of a trajectory, gains of 2.348, 3.331 and 5.326 bits at $m = 16$, 64, 1024, against $\frac{1}{2}\log\frac{\pi m}{2} = 2.326$, 3.326, 5.326, with shares 0.15, 0.052 and 0.0052 (more in Appendix A). So the share of the raw loss that the law saves tends to 0, but its absolute gain per step equals its value for the corresponding question: the share vanishes because of micro-detail that no theory addresses. Both tasks learn the law at the same price, so the theory is "in" the raw predictor. The tasks differ in their *target*, which losses are counted, not in what must be learned or what the law is worth per step; and scoring questions is right when what we want is answers. Part (b) holds by construction, since the toy asks only about the conserved quantity.

Proposition 1.5 (Raw modelling plus an extractor)**.** *Let μ be a computable probability measure on $\mathbb{X}^{\mathbb{N}}$ for a finite alphabet $\mathbb{X}$ (the raw world), $(\varphi_j)_{j\in\mathbb{N}}$ a computable family of sentences, and η a partial computable function (the* extractor*) such that $\eta(j,x) \in \{0,1\}$, when defined, is computed from a finite prefix of x. Assume that for μ-almost every x the labelling $f_x : \varphi_j \mapsto \eta(j,x)$ is consistent with B, and put $\mu_\eta(D) := \mu\{x : f_x \supseteq D\}$. Then for every finite data set D, $M_A(D) \geq 2^{-K(\mu)-K(\eta)-c}\mu_\eta(D)$, so $L_A(D) \leq -\log \mu_\eta(D) + K(\mu) + K(\eta) + c$. In particular, for a computable sequence x with f_x consistent with B, $L_A(D) \leq K(x) + K(\eta) + c$ whenever $f_x \supseteq D$.*

A theory machine samples a raw sequence with law μ from its tape, runs the extractor and emits the labelled sentences; **??** does the rest (Appendix A). The extractor is the bridge from raw data to the question language, and its cost $K(\eta)$ is not small in real cases: it contains whatever recognises a planet in an image. The converse fails badly, since answers do not determine raw data.

Proposition 1.6 (Answering and predicting)**.** *(a) If a consistent r.e. theory T proves every datum of a consistent reveal sequence, the cumulative loss in the semimeasure convention is at most $-\log \mathbb{U}_A(\mathrm{Ext}T) \leq K_{\equiv}(T) + c$, in any order (**??**). (b) On the lottery stream of* **??***, M_A has bounded loss, while no consistent r.e. theory proves every datum and $W_A(D_n) \to 0$. (c) If T^* is consistent and r.e., $B \cup T^*$ is incomplete and two-valued data have $\mathrm{Cn}(D^{\pm}_{\infty}) = \mathrm{Cn}(T^*)$, the limit posterior is the prior restricted to the consistent extensions of T^*, and on every sentence independent of $B \cup T^*$ the limits of* Bel *and* Dis *lie in* $(0,1)$ *and that of* Indep *is positive; three-valued data giving every sentence its T^*-trichotomy label identify $[T^*]$ (**??**). (d) If $M_A(D) > 0$, then $\mathrm{Bel}_D(\bigwedge_{i\leq k}\varphi_i) \geq 1 - \sum_{i\leq k}(1-\mathrm{Bel}_D(\varphi_i))$, and $\mathrm{Bel}_D(\psi) \geq \mathrm{Bel}_D(\varphi)$ whenever $B \cup \{\varphi\} \vdash \psi$.*

Parts (a)–(c) are **??????**; (d) holds because a theory that proves two sentences proves their conjunction (Appendix A). So answering is the stronger achievement ((a) one way, (b) not the

other); predicting a presentation of data does not yield answers to questions not asked (c); and the beliefs of SAI cohere (d), which a predictor of separate labels does not: in the semimeasure convention consistent SOI never becomes certain of an entailed or unasked sentence (**??**).

The objection from statistical practice. Much of physics confronts theory with near-raw data through generative models and information-theoretic losses: gravitational-wave searches match waveform templates against the strain series with a Gaussian noise likelihood, cosmology fits likelihoods of microwave-background data, particle physics simulates detectors event by event. We think D survives in comparative form, because these pipelines exist to answer a few questions (a mass, a spin, a cosmological parameter, whether a particle exists). But the generative model is theory plus instrument plus noise, and the proof-requiring core of SAI, with no native noise model, sits far from this practice (**??**).

Purpose and logarithmic loss. Grading a prediction for use and scoring credences are different jobs. For the first, the purpose decides what is graded and with which tolerance, as with the interval data of **??**. For the second, every strictly proper score is optimised by the true distribution, Bayesian updating compares theories by cumulative logarithmic loss, and the logarithmic score is, up to positive affine transformation, the only strictly proper local scoring rule on more than two outcomes (a result attributed to Bernardo, 1979); yet which proper score is privileged for epistemic purposes is contested: accuracy-first arguments favour quadratic inaccuracy measures (Leitgeb and Pettigrew, 2010; Pettigrew, 2016), and conditionalisation maximises expected epistemic utility under general conditions (Greaves and Wallace, 2006). For bounded or misspecified models different proper losses select different best models, and logarithmic loss on raw data spends capacity on detail no purpose asks for. Solomonoff axiom induction lets the purpose fix the propositions and scores credences on those; even then the price of a datum can depend on the loss convention by an unbounded amount (**??**). We do not claim that the logarithmic score is privileged.

The verdict on D. D holds in comparative form. Its formal content is a change of data type, logical coherence, partiality and the choice of target. It is not a new inference principle, since SAI predicts truth values of sentences and one-tape SAI $\asymp$ one-tape consistent SOI (**??**). The real difficulty of moving from raw data to questions is the extractor, which SAI takes as given.

1.4 What a theory is, after Craig

The idealised formalism cannot see shape. Every consistent assigner of truth values is an axiom set at $O(1)$ extra bits: coin-free SAI $\asymp$ coin-free consistent SOI (**??**, Craig's construction inside the universal prior) and one-tape SAI $\asymp$ one-tape consistent SOI (**??**). Adding vocabulary changes old-vocabulary weights by $O(1)$ (**????**); coin-free, an r.e. axiom set is a finite theory in a three-sorted extension, up to $O(1)$ (**??**). Program length sees neither the reasoner nor the vocabulary (**??**); the posterior cannot move between deductively equivalent formulations, which in the flagpole toys below even differ in complexity by $O(1)$ whatever the number of laws (Proposition 1.8). The diagonal separations from SOI under time bounds come from free deduction, not from axioms (**??**). So the idealised formalism cannot see the difference between a theory and a program. Similarly, Khardon and Roth (1997) give a learning-to-reason algorithm for a class of propositional languages that has no efficient reasoning algorithm when represented as a formula-based knowledge base: competence at answering queries need not go through an

explicit theory. So D does not by itself support S as a claim about what an ideal inducer should produce.

Where shape matters, it is relative to a reasoner and a vocabulary, and only when the written theory is charged: in the model of **??** the exponential separations are between pairs (reasoner, vocabulary) (**??**), and declarative against procedural form matters only by a constant factor once intermediate concepts are available and the reasoner can chain through them (**??**); in the settings of **??** what pays is theoretical relations and entities, not vocabulary (**??**). Nothing in the formal results favours sentences over classes of models as what a theory is; and since the limit posterior can live on random or false theories (**????**), nothing realist should be said about induced theories.

1.5 Explanation

In the axiom view, T explains φ when $B \cup T \vdash \varphi$, the deductive-nomological pattern (Hempel and Oppenheim, 1948), and an explanation is better the more a short theory derives, as in unification accounts (Friedman, 1974; Kitcher, 1981); the explanatory power of **??** (bits of data explained per bit of axiom) makes this precise and is related to Myrvold's Bayesian account of unification (Myrvold, 2003) in **??**. But derivability is symmetric where explanation is not: a pole's height and the sun's elevation entail its shadow's length, and the shadow's length and the elevation entail the height (Bromberger's example, Bromberger, 1966, as attributed by Glymour, 1978). Take poles and reals, with B the theory of real closed fields, pole constants $p_1, \dots, p_{N_\mathrm{p}}$, heights $\mathrm{ht}(p)$, and shadow lengths $\mathrm{sh}(p,u)$ when the tangent of the sun's elevation is $u > 0$. The shadow law has two formulations,

$$\Lambda_h := \forall p\, \forall u\, \big(u > 0 \to \mathrm{sh}(p,u) \cdot u = \mathrm{ht}(p)\big), \qquad \Lambda_s := \forall p\, \forall u\, \big(u > 0 \to \mathrm{sh}(p,u) \cdot u = \mathrm{sh}(p,1)\big) \wedge \forall p\, \mathrm{ht}(p) = \mathrm{sh}(p,1),$$

each of which can be combined with initial conditions on the heights or on the shadows at $u = 1$. A derivation is charged by the number of instances of the law's universally quantified conjuncts it uses. Appendix A states the next two propositions precisely.

Proposition 1.7 (The flagpole)**.** *(a) The four theories so obtained are deductively equivalent over B, so they lie in one deductive class: no data set separates them, and in the coin-free version the posterior odds of any two programs for them equal their prior odds on every data set. (b) Under derivation charges, on every shadow datum with $u \neq 1$ the two theories whose initial conditions sit at the hub of their formulation (heights with Λ_h, shadows at $u = 1$ with Λ_s) need one instance and the other two need two; the hub is a property of the formulation, not of the world, and the texts of the two hub theories differ by $|\Lambda_s| - |\Lambda_h|$, independently of N_p and the data. (c) Two causal extensions with intervention vocabulary, one with the height and one with the shadow exogenous, have the same consequences in the old vocabulary, and one interventional datum separates them.*

Kitcher's unificationist answer to the asymmetry, as we read it (Kitcher, 1989), is that the best systematisation of all our beliefs derives the dimensions of objects from their origin and development, so that a height-from-shadow pattern would add a pattern, and one that fails for objects that cast no shadow. To represent it, add k further laws $\forall p\, \mathrm{g}_j(p) = \theta_j \cdot \mathrm{ht}(p)$ (or, in shadow form, with $\mathrm{sh}(p,1)$ in place of $\mathrm{ht}(p)$), and compare the height hub (Λ_h, laws in height form, heights given), the mixed theory (Λ_h, height form, shadows given) and the shadow hub (Λ_s, shadow form, shadows given), also when some poles cast no shadow.

Proposition 1.8 (The flagpole with a system of laws)**.** *(a) The three theories are deductively equivalent, so for every k the posterior odds of their classes never move. (b) Under the universal*

prior the two hubs differ in complexity by at most a constant independent of k, of the number of poles and of the numerals. (c) In a literal code without definitions the shadow hub is longer by $(|\Lambda_s| - |\Lambda_h|) + (k + N_{\mathrm{p}})\Delta_{\mathrm{sh}}$, where Δ_{sh} is the extra length of the term $\mathrm{sh}(x, 1)$ over $\mathrm{ht}(x)$ (zero with a primitive for the shadow at $u = 1$); when some poles cast no shadow it is longer by at least a constant times k, even if $\Delta_{\mathrm{sh}} = 0$; with one definition the overhead is a constant. (d) Under derivation charges the two hubs need one instance on each datum about a g_j, and the mixed theory needs two.

What the toys show. The explanation relation of SAI is direction-free, and its posterior cannot move between deductively equivalent theories. Kitcher's answer is a whole-theory simplicity effect, which SAI reproduces only under a listing code that charges each law statement separately; under the universal prior it is $O(1)$, as program length makes the reasoner invisible (**??**), and under derivation charges only the mixed theory loses. Direction enters cleanly with interventional questions, once the base language can pose them; but the intervention semantics was written into the causal theories by hand, so the toy is consistent with the interventionist account (Woodward, 2003) and gives it no support over rivals such as the causal-process account (Salmon, 1984). We conclude that derivation from a short theory is explanation in the unification sense, with at least four known gaps, and we do not claim that derivation is explanation outright. The gaps are asymmetry, as above; explanation by approximate correction, since Newton's theory does not entail Kepler's laws but, in an exact toy, proves the honest old law on every window and refutes the exact one (**??**), so that "T explains φ" must include "T derives an honest approximation of φ with an error bound on a domain"; irrelevance, as in the standard examples of hexed salt and of a man taking birth-control pills, where the length prior penalises an irrelevant axiom in a whole theory but a derivation does not mark which axioms it used; and the conjunction problem (L is derived from $L \wedge L'$), which the length prior and the explanatory power address in part. Unification can also track an explanatory gain without a gain in accuracy: against Ptolemy's system with the coincidences of retrogression and opposition stated as axioms, which has the same fit, Copernicus's wins exactly when the heliocentric reorganisation costs fewer bits than the coincidences (**??**).

1.6 What separates SAI from raw-data prediction

We replace claim P by the formal differences between SAI and a predictor of an observation stream, sorted into advantages for claims D and E, neutral differences and liabilities (Table 2).What does not separate the two is the inference principle, a Bayes mixture over a universal class, and, in the idealised setting, the shape of hypotheses (Section 1.4). This is the form of the progress claim that we believe, liabilities included: for what a theory does, SAI is a step in the right direction; for what a theory is, the idealised formalism is neutral.

A Proofs for what a theory does and is

(To be written.)

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	difference	support
Advantages for claims D and E		
1	*Data type:* labelled sentences of a chosen language; questions are sets of sentences.	**??** and Def. 1.2
2	*Logical coherence:* every consequence of the data gets posterior probability 1, while in the semimeasure convention consistent SOI is never certain of an unasked consequence; Bel is closed under conjunction up to additive error and monotone under entailment.	**??**; Prop. 1.6(d)
3	*Partiality:* "independent" is a native third value, whose mass is the semimeasure defect; for a k-way question the answer masses and the unanswered mass sum to 1.	**??** and Prop. 1.3
4	*An observed "independent" label* separates theories from labellers by an unbounded amount on one datum, coin-free and one-tape, when $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and $\mathsf{Q} \subseteq \mathrm{Cn}(B)$.	**??**
5	*Laws are sentences with belief values:* $\mathrm{Bel}_D(\forall x\, \varphi)$ differs from the probability that all future data conform.	**??**
6	*Answering questions costs no more description than raw modelling plus an extractor,* and can cost far less.	Props. 1.4 and 1.5
Neutral differences		
7	*Piecemeal confirmation:* credence in individual laws, although only whole theories meet the data. This is the standard Bayesian treatment of Duhem's problem (Dorling, 1979), which Strevens (2001) argues begs the question; SAI inherits the debate, and its allocation of blame is fixed by the prior, hence by the syntax.	definitions; an argument
8	*Truth from unprovability:* if $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, belief or independent mass on a Π_1 sentence certifies its truth; a curiosity of arithmetic data.	**??**
9	*Explanatory direction needs questions the language can pose:* interventional data separate causal theories that the old vocabulary cannot.	the toy Prop. 1.7(c); in general an argument
Liabilities		
10	*Random theories, false theories:* on the lottery stream bounded one-tape loss is achieved by random theories and by no consistent r.e. theory; in true arithmetic with true data the limit posterior can live on false theories (a possibility result, with adversarial data).	**????**
11	*Simplicity is relative to the language, within bounds:* faithful translations change weights by $2^{\pm(K(\tau)+c)}$; within that bound the base language decides real choices, such as the preferred compatible orbit of Mars on noise-free interval data.	**??**; **??**

Table 2: What formally separates SAI from raw-data prediction. Items 7 and 9 and the sorting are arguments; the other items are results.

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